22 **C** is a circle with centre the origin.

A tangent to **C** passes through the points (–20, 0) and (0, 10)

Work out an equation of **C**.
You must show all your working.

Tangent gradient = (10 – 0)/(0 – (–20)) = 1/2.

A radius to the touching point is perpendicular: gradient –2.

Tangent: $y = x/2 + 10$; radius from origin: $y = -2x$.

At the contact point: $-2x = x/2 + 10$, so $x = -4$.

Then $y = 8$, radius$^2 = (-4)^2 + 8^2 = 80$.

Circle C: $x^2 + y^2 = 80$.

5 marks total
Tangent gradient 1/2
Radius gradient –2
Two line equations
Contact (–4,8)
$x^2+y^2=80$

(Total for Question 22 is 5 marks)

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